

TrES-5b

R-0824

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_181210 · created 2026-07-28T15:46:51+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458462.5941 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.106 +/- 0.045
Transit depth (Rp/Rs)^2	1.12 +/- 0.94 [%]
Orbital Inclination (inc)	84.91 +/- 2.85
Airmass coefficient 1 (a1)	1.016 +/- 0.017
Airmass coefficient 2 (a2)	-0.007 +/- 0.011
Scatter in the residuals of the lightcurve fit is	1.66 %
Best Comparison Star	#1 - [385, 160]
Optimal Method	PSF photometry
Transit Duration (day)	0.045 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.106 ± 0.045 vs 0.142 (deviation 0.8σ)
Duration fit vs published	0.045 d vs 0.0758 d (deviation 2.8σ)
Mid-time O-C	6.5 min
Depth SNR	2.4
Residual scatter	1.66 %

Light curve

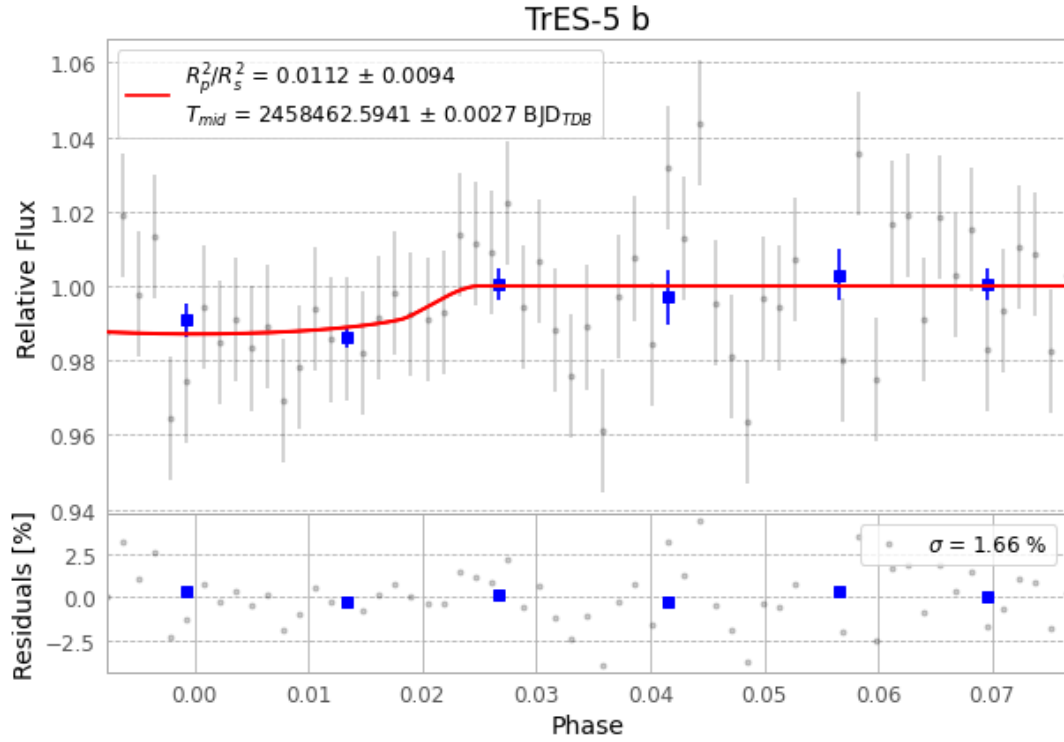


Figure 1 — FinalLightCurve_TrES-5 b_2018-12-09.png (SHA-256 5860cf198e3e4e1d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [301, 404], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0e1c5b50f80f5cb2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

61 FITS frames (40 MB), TRES-5181210015712.FITS → TRES-5181210045712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-181210.log (7b657f17ee48...), FinalLightCurve_TrES-5 b_2018-12-09.png (5860cf198e3e...), FinalLightCurve_TrES-5 b_2018-12-09.csv (712c5394cbe7...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 15:46Z.